Exp2

23/8/24

# Study of Different types of Network Cables.

**Aim:-** To study different types of Network Cables.

Different types of Cables used in networking are:

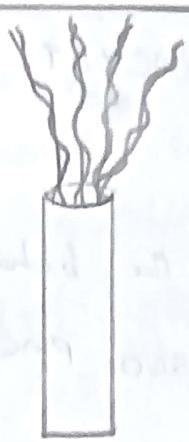
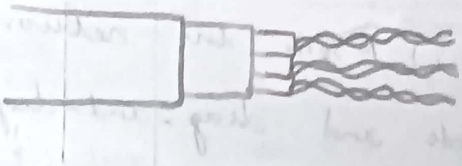
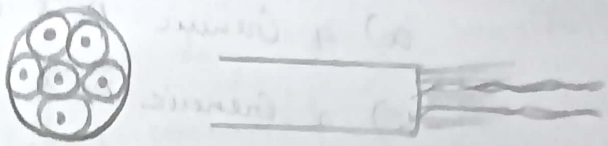
1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre optic Cable.

| Cable Type | Category | Maximum Data Transmission | Advantage/ Disadvantages | Application/ Use |
|---|---|---|---|---|
| UTP | Category 3 | 10 bps | **Advantages**<br>* Cheaper<br>* Easy to install as they have smaller diameter<br>**Disadvantage**<br>* More prone to (EMI) Electromagnetic Interference and noise | 10 Base-T Ethernet |
| | Category 5 | upto 100 Mbps | | Fast Ethernet, Gigabit Ethernet |
| | Category 5e | 1Gbps | | |

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|---|---|---|---|---|
| STP | Category 6, 6a | 10 Gbps | Advantages<br>* Shielded<br>* Faster than UTP<br>* Less susceptible to noise + interference<br>Disadvantages:<br>* Expensive<br>* Greater installation effort | Gigabit Ethernet, 10G Ethernet (55m)<br>Widely used in data centres |
| SSTP | Category 7 | 10 Gbps | | Gigbit Ethernet<br>10G Ethernet (100m) |
| Coaxial Cable | RG-6<br>RG-59<br>RG-11 | 10-100 Mbps | Advantages:<br>* High bandwidth<br>* Immune to interference<br>* Low Loss bandwidth<br>Disadvantages:<br>* Limited distance<br>* Cost<br>* Size is bulky | Speed of signal is 500m<br>Television network<br>High speed internet connections |
| Fibre optics Cable | Single Mode<br>Multi Mode | 100Gbps | Advantages:<br>* High Speed<br>* High bandwidth<br>* Long distance<br>Disadvantages:<br>* Expensive<br>* Requires skilled installers | Maximum distance of fibre optics Cable is around 100 metres |

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| | |
|---|---|
| UTP |  |
| STP<br><br>SSTP |  <br>  |
| Coaxial Cable |  |
| Fibre optics cable |  |

Result:- Thus the study of different types of Network Cables is done.

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